

Reproducible Manuscript Outputs

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Purpose

This R Markdown file documents and reproduces the manuscript analysis for estimating waking sedentary time in NHANES 2011-2014 wrist accelerometer data. The project combines posture predictions from the CHAP wrist model with sleep/nonwear information from SWaN to estimate each participant's usual daily waking sedentary time.

The main goal is to produce survey-weighted, nationally representative estimates of waking sedentary time after excluding sleep/nonwear periods. The analysis first calibrates a sleep/nonwear percentage cutoff against NHANES self-reported sleep duration, then applies the selected cutoff at the native analysis epoch, summarizes sedentary time to the participant level, and estimates overall and subgroup means using the NHANES survey design.

This file is also designed as a reproducibility driver. It reads the analysis-ready accelerometer data, derives the analytic sample, recalibrates the sleep cutoff for the selected epoch resolution, generates tables and figures, and writes manuscript-ready outputs to `outputs/` and `figures/manuscript/`.

The analysis can be rendered using either the 1-hour or 30-minute epoch data. Demographic and NHANES survey design variables are taken from `data/analysis_ready_1h_epoch/nosleep_data.csv.gz` for both

epoch choices; the hip-based sedentary values in that file are used only as a metadata source and are not used as the primary sedentary behavior measure.

Packages

Analysis Configuration

This render uses the 1h epoch dataset. Sleep/nonwear removal is applied at the native epoch level before participant-day sedentary totals are calculated. For figures that display hourly patterns, the native epoch records are aggregated to hour only after the sleep/nonwear rule has been applied.

The primary analytic unit is the NHANES participant. Epoch-level accelerometer records are used to construct participant-day and participant-level sedentary time summaries, but survey-weighted estimation is performed at the participant level to avoid giving participants with more records or more valid days disproportionate influence.

Load And Prepare Data

Reusable Functions

Table 1

Analytic sample and wear-day distribution by NHANES cycle (2011-2014).

Table 1: Analytic sample and wear-day distribution by NHANES cycle (2011-2014).

Full days	2011-2012	2013-2014	Total (2011-14)
1	29 (0.42)	29 (0.38)	58 (0.40)
2	37 (0.54)	31 (0.40)	68 (0.47)
3	48 (0.70)	30 (0.39)	78 (0.54)
4	37 (0.54)	31 (0.40)	68 (0.47)
5	56 (0.82)	30 (0.39)	86 (0.59)
6	68 (0.99)	43 (0.56)	111 (0.76)
7	6,568 (95.98)	7,497 (97.48)	14,065 (96.77)
Sample Analyzed (1-7)	6,843 (100.00)	7,691 (100.00)	14,534 (100.00)

Notes: A full day is a 24-hour day represented by 24 hourly records; partial initiation/return days are excluded.

Cutoff Sensitivity Grid

This step does not select the final sleep/non-wear cutoff. It evaluates a coarse grid of candidate cutoffs in 5-percentage-point increments to summarize how survey-weighted sleep duration and waking sedentary time change across plausible decision rules. The resulting `sensitivity_results` object is used for the cutoff-sensitivity display in Figure 4. The final cutoff is selected in the next step using a finer objective curve.

Optimal Sleep Cutoff Selection

The final sleep/non-wear cutoff is selected from a fine-resolution objective curve. For each candidate cutoff, the analysis compares survey-weighted accelerometer-derived sleep duration with the survey-weighted NHANES self-reported sleep benchmark. The selected cutoff is the value that minimizes the absolute difference between those two sleep-duration estimates. This fine-resolution selected cutoff, not the coarse sensitivity-grid cutoff, is used for the primary waking sedentary-time analysis.

Objective Curve for Cutoff Selection

The fine-resolution objective curve selected a sleep/non-wear cutoff of 90.9%. At this cutoff, the survey-weighted accelerometer-derived sleep estimate was 6.92 hours/day, and the absolute difference from the survey-weighted self-reported sleep benchmark was 0.033 hours/day. This selected value is used for the primary waking sedentary-time analysis.

Displaying the objective function plot to visualize the cutoff selection process

Main Analytic Dataset At Selected Cutoff

Main Results

Using the selected sleep/nonwear cutoff of 90.9%, the final analytic sample included 14,534 participants. The survey-weighted mean waking sedentary time was 11.33 hours/day (SE = 0.04; 95% CI: 11.25, 11.42). Mean waking time after sleep/nonwear exclusion was 17.08 hours/day, and mean sleep/nonwear time was 6.92 hours/day.

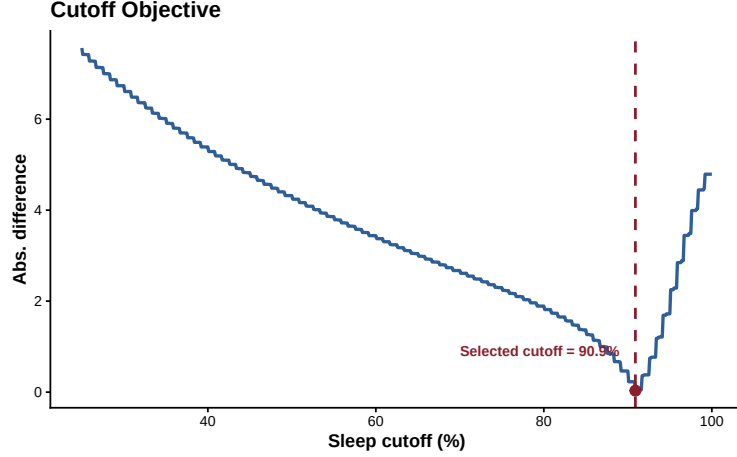


Figure 1: Objective function for sleep-cutoff calibration. The curve shows the absolute difference between survey-weighted accelerometer-derived sleep duration and survey-weighted self-reported sleep duration across candidate sleep-percentage cutoffs. The dashed vertical line marks the selected cutoff.

Table 2: Main survey-weighted estimates at the selected sleep/nonwear cutoff.

Measure	Mean (h/day)	SE	95% CI lower	95% CI upper
Waking sedentary time	11.33	0.04	11.25	11.42
Waking time	17.08	0.05	16.98	17.18
Sleep/nonwear time	6.92	0.05	6.82	7.02

Table 3: Main survey-weighted estimates at the selected sleep/nonwear cutoff, overall and by NHANES cycle.

Analysis group	Measure	Mean (h/day)	SE	95% CI lower	95% CI upper
Pooled	Waking sedentary time	11.33	0.04	11.25	11.42
	Waking time	17.08	0.05	16.98	17.18
	Sleep/nonwear time	6.92	0.05	6.82	7.02
Cycle 2011-2012	Waking sedentary time	11.44	0.06	11.32	11.56
	Waking time	17.26	0.07	17.13	17.39
	Sleep/nonwear time	6.74	0.07	6.61	6.87
Cycle 2013-2014	Waking sedentary time	11.23	0.06	11.10	11.36
	Waking time	16.91	0.08	16.74	17.07
	Sleep/nonwear time	7.09	0.08	6.93	7.26

Justification for Combining Survey Cycles

Table 4: Survey-weighted mean sedentary time by NHANES cycle.

Dataset	Mean sedentary time (hours/day)	SE
2011-2012	11.44	0.06
2013-2014	11.23	0.06

Justification for Combining Survey Cycles

To assess whether the two NHANES cycles could be pooled, we compared survey-weighted estimates of mean sedentary time between 2011-2012 and 2013-2014. The mean sedentary time was 11.44 hours/day (SE = 0.06) in 2011-2012 and 11.23 hours/day (SE = 0.06) in 2013-2014. The survey-weighted regression estimate of the difference was -0.21 hours/day. This difference is small relative to the overall mean and provides no evidence of a meaningful change in sedentary time between cycles. Based on this, the two cycles were combined for all subsequent analyses. This approach is consistent with NHANES analytic guidance, which supports pooling survey cycles when key estimates are stable across time (NCHS, 2026). The survey-weighted regression estimate of the difference was -0.21 hours/day ($p = 0.0222$).

NHANES Analytic Approach

The primary estimand in this manuscript is the population mean of each participant’s usual daily waking sedentary time. The analysis therefore treats the NHANES sample person, not the hour, as the survey-weighted analytic unit. This matches NHANES guidance: sample weights are assigned to sample persons and account for unequal probability of selection, nonresponse, and post-stratification to U.S. population control totals; these weights are needed to produce unbiased national estimates from continuous NHANES (NCHS Sample Design Module).

The repeated accelerometer summary records were still used fully to derive the participant-level exposure. For each candidate sleep cutoff, SWaN sleep/nonwear percentages determined whether an epoch was treated as waking time, CHAP sitting probabilities contributed fractional sedentary hours using the epoch duration, daily totals were computed, and valid days were averaged within participant. The person-level mean is therefore a summary of all available valid epoch- and daily-level information for that participant. What is intentionally removed before the main survey estimation is the repeated-record structure itself, because leaving epochs or days as separate weighted records would give participants with more observed records or valid days more influence than participants with fewer records. That would change the estimand from a population mean across people to a record-weighted mean across observed person-epochs or person-days.

The manuscript addresses NHANES analytic guidance in four specific ways. First, because the analysis combines NHANES 2011-2012 and 2013-2014, the examination weight is divided by two to form a 2011-2014 weight, consistent with NHANES instructions that combined weights from 2001-2002 onward are created by dividing the 2-year weights by the number of 2-year cycles (NCHS Weighting Module). Second, the survey design includes the masked variance strata and masked PSU with `nest = TRUE`, which follows NHANES guidance for public-use design variables in continuous NHANES. Third, standard errors and confidence intervals are estimated with Taylor series linearization through the R `survey` package, the variance-estimation approach recommended by NCHS for NHANES analyses (NCHS Variance Estimation Module). Fourth, subgroup estimates are obtained from the survey design object using domain/subset analyses rather than by defining a new design after dropping non-subgroup records, preserving the design information needed for valid subgroup standard errors.

Thus, the person-level summary does not discard the epoch- or day-level information used to estimate sedentary behavior; it uses those data to construct the participant’s daily mean and then applies the NHANES person-level survey design to the quantity that each sampled person represents. Epoch-level or day-level mixed models can be useful as sensitivity analyses for within-person variability, but they answer a different modeling question and require additional assumptions about the repeated-measure correlation structure and how person-level NHANES weights should be represented across repeated observations.

Demographic Variables and Survey Weights

Demographic and anthropometric variables were taken from the NHANES metadata file and reduced to one record per participant before survey estimation. Age was grouped as <18, 18-34, 35-44, 45-64, and ≥65 years. Race/ethnicity was retained using the NHANES-coded categories available in the analysis data. BMI was calculated from measured weight and height as weight in kilograms divided by height in meters squared, then grouped as under/normal weight (BMI < 25), overweight (25 ≤ BMI < 30), and obesity (BMI ≥ 30). Participants with missing age, race/ethnicity, BMI, design variables, or weights were not recoded

into another category; missing subgroup information remains missing and is excluded only from the relevant subgroup/domain estimate.

These variables are used to describe population heterogeneity in waking sedentary time, not as repeated epoch-level variables. Subgroup estimates for age, gender, race/ethnicity, and BMI are calculated from the participant-level survey design object using domain analyses. This preserves the full NHANES design information for standard error estimation while estimating the mean for the population domain of interest. Survey weights are treated as design weights rather than ordinary covariates: the recalibrated examination weight defines each participant’s contribution to national estimates, while masked strata and PSU variables define the variance structure.

Table 5: Missingness of demographic and survey design variables relative to the final analytic sample.

Variable	Analytic sample N	Missing n	Missing %	Available n
Age group	14534	0	0.00	14534
BMI category	14534	134	0.92	14400
Education	14534	0	0.00	14534
Gender	14534	0	0.00	14534
Marital status	14534	0	0.00	14534
Masked PSU	14534	0	0.00	14534
Masked stratum	14534	0	0.00	14534
Race/ethnicity	14534	0	0.00	14534
Recalibrated examination weight	14534	0	0.00	14534

The final analytic sample included 14,534 participants after applying accelerometer-data availability and survey-design requirements. Missingness in demographic and survey design variables is shown relative to this analytic denominator so subgroup tables can be interpreted with the correct available sample for each domain.

Overview of Sedentary-Time Estimation

The analysis proceeded in the following steps:

1. **Start with wrist accelerometer data.** We used NHANES 2011-2012 and 2013-2014 wrist accelerometer files. Each file records movement over time for one participant, usually across several wear days.
2. **Prepare the files for prediction models.** The raw files were organized by participant and day and converted into the formats needed by the posture and sleep/wear models.
3. **Identify sitting in 10-second epochs.** The CHAP/DeepPostures model classified each 10-second epoch as sitting or not sitting. For example, one minute contains six 10-second epochs, so a participant could have six sitting/not-sitting predictions in a single minute.
4. **Identify sleep and non-wear in 30-second epochs.** The SWaN model classified each 30-second epoch as wake wear, sleep, or non-wear. For example, one hour contains 120 30-second epochs, and those epochs were used to describe how much of the hour was sleep or non-wear.
5. **Summarize predictions for analysis.** The 10-second sitting predictions and 30-second sleep/non-wear predictions were aggregated into new hourly epoch-level summaries for each participant-day. For example, a 1-hour analysis epoch contains 360 10-second sitting predictions and 120 30-second sleep/non-wear predictions; these were summarized as the percentage of the epoch classified as sitting and the percentage classified as sleep/non-wear.
6. **Choose the sleep/non-wear cutoff.** Candidate cutoffs were evaluated by comparing survey-weighted accelerometer-derived sleep duration with NHANES self-reported sleep duration. The cutoff with the

smallest absolute difference was selected for the primary waking sedentary-time analysis. This cutoff is a practical rule for classifying accelerometer summaries, not an exact biological boundary.

7. **Keep only waking epochs for sedentary-time estimation.** Epochs below the selected sleep/non-wear cutoff were treated as waking epochs. Epochs at or above the cutoff were removed from waking sedentary-time estimation.
8. **Calculate sedentary hours for each day.** For each participant-day, sitting percentages were multiplied by epoch duration and added across waking epochs. The sum across waking epochs gave total waking sedentary time for that day.
9. **Average days within each participant.** Daily waking sedentary-time values were averaged across each participant's valid wear days. For example, if a participant had four valid days, the four daily values were averaged to estimate that participant's usual daily waking sedentary time.
10. **Use the NHANES survey design.** Each participant's usual daily waking sedentary time was analyzed using NHANES examination weights, masked variance strata, and masked primary sampling units. The 2011-2012 and 2013-2014 cycles were combined using recalibrated 4-year weights.
11. **Estimate national and subgroup values.** Survey-weighted means and confidence intervals were estimated for the full population and for subgroups defined by age, gender, race/ethnicity, and BMI category.

Figure 1

Workflow for deriving hourly sitting and sleep summaries. (A) Raw 80 Hz accelerometer signal. (B) CHAP sitting predictions and SWaN sleep/wake/non-wear classifications. (C) Hourly sitting and sleep/non-wear percentages used to define waking sedentary time under candidate sleep cutoffs.

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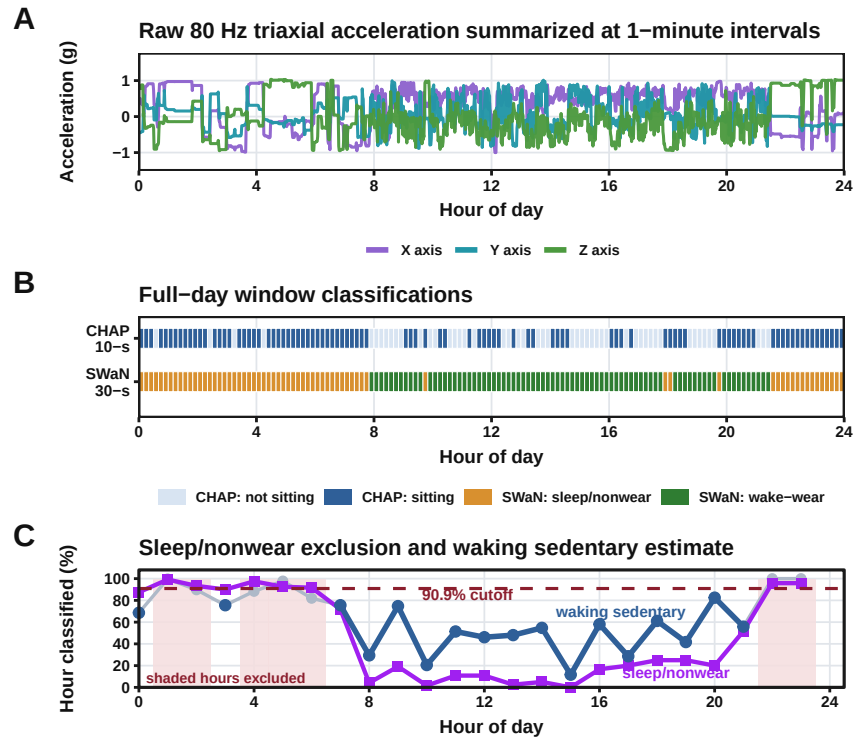


Figure 2: Workflow for deriving hourly sitting and sleep summaries. (A) Raw 80 Hz accelerometer signal. (B) CHAP sitting predictions and SWaN sleep/wake/non-wear classifications. (C) Hourly sitting and sleep/non-wear percentages used to define waking sedentary time under candidate sleep cutoffs.

Table 2

Comparison of sedentary time estimates across studies by measurement method, datasets, and population.

Table 6: Comparison of sedentary time estimates across studies by measurement method, datasets, and population.

Paper	Dataset & Population	Method	Estimate & Notes
This study	NHANES 2011-2014, wrist-worn 80 Hz accelerometer data; 14,534 participants, aged 3+ years.	CHAP wrist model combined with SWaN.	11.33 hours/day. Estimated waking sedentary time. Mean waking time was 17.08 hours/day after sleep exclusion.
Matthews et al. (2008)	NHANES 2003-2004; hip-worn ActiGraph data; 6,329 participants, aged 6+ years.	Intensity threshold of <100 counts/min.	7.70 hours/day. Represented 54.90% of monitored time. Mean monitor wear time was 13.90 hours/day.
Yang et al. (2019)	NHANES 2001-2016; 51,896 children and adolescents.	Self-reported total sitting time from questionnaire.	~6.40 hours/day in adults; 8.20 hours/day in adolescents. Sitting time increased in both groups from 2001 onward.
Ussery et al. (2018)	NHANES 2015-2016; 5,923 U.S. adults aged 18+ years.	Self-reported sitting time on a typical day.	~6.40 hours/day. Excludes sleeping time; 25.70% of adults reported sitting for more than 8 hours per day.

Figure 2

Survey-weighted hourly patterns of wrist-derived sedentary behavior by day type before and after sleep exclusion. Panel A shows CHAP-predicted sitting time including sleep periods; Panel B shows waking sedentary time after excluding records with sleep/nonwear greater than or equal to the selected 90.9% cutoff. Lines represent hourly means and shaded bands indicate 95% confidence intervals.

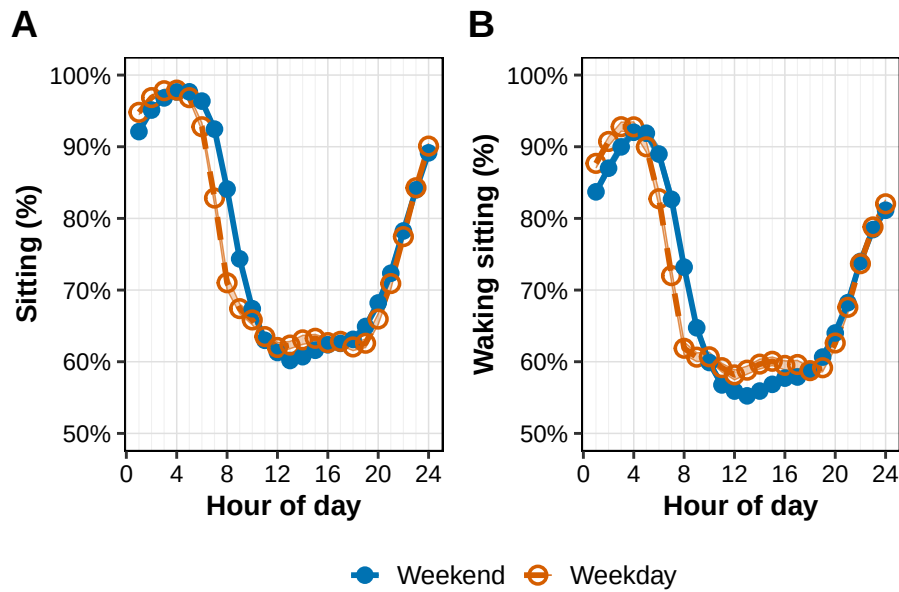


Figure 3: Survey-weighted hourly patterns of wrist-derived sedentary behavior by day type before and after sleep exclusion. Panel A shows CHAP-predicted sitting time including sleep periods; Panel B shows waking sedentary time after excluding records with sleep/nonwear greater than or equal to the selected 90.9% cutoff. Lines represent hourly means and shaded bands indicate 95% confidence intervals.

Figure 3

Distribution of participants by mean daily waking sedentary time and corresponding 24-hour behavioral composition. Panel A shows the proportion and unweighted number of participants in each sedentary-time category. Panel B shows the mean daily allocation to sedentary time, sleep/nonwear, and other waking time within each category.

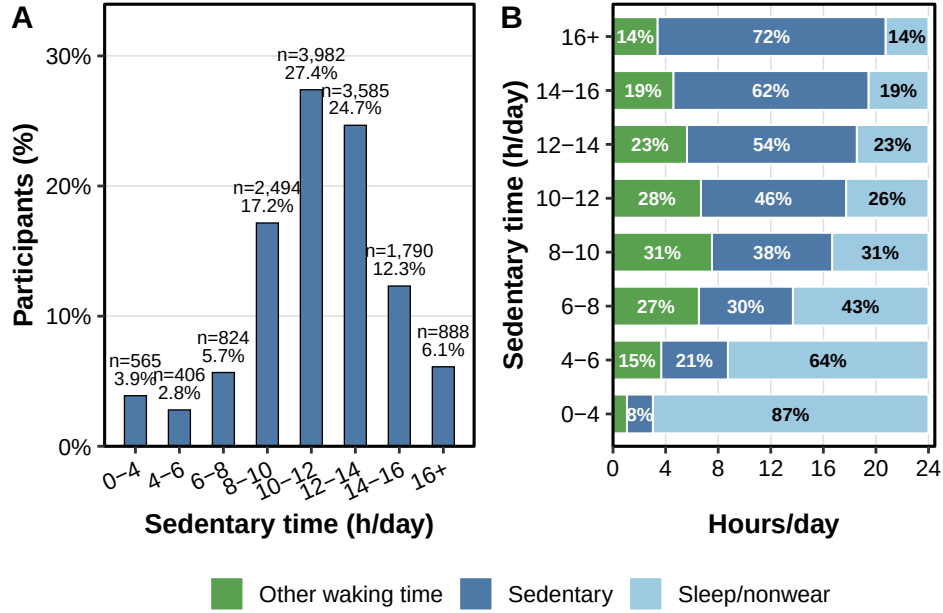


Figure 4: Distribution of participants by mean daily waking sedentary time and corresponding 24-hour behavioral composition. Panel A shows the proportion and unweighted number of participants in each sedentary-time category. Panel B shows the mean daily allocation to sedentary time, sleep/nonwear, and other waking time within each category.

Table 3

Weighted amount of time spent in sedentary behavior (mean hours/day), by gender and age, United States, 2011-2014.

Table 7: Weighted amount of time spent in sedentary behavior (mean hours/day), by gender and age, United States, 2011-2014.

Age group (years)	Male N (%)	Male Mean (95% CI)	Female N (%)	Female Mean (95% CI)
All ages	7,099 (48.84%)	11.47 (11.33, 11.60)	7,435 (51.16%)	11.21 (11.10, 11.32)
<18	2,331 (16.04%)	10.53 (10.35, 10.72)	2,244 (15.44%)	10.19 (10.02, 10.35)
18-34	1,446 (9.95%)	10.86 (10.60, 11.13)	1,506 (10.36%)	10.59 (10.32, 10.86)
35-44	751 (5.17%)	11.07 (10.68, 11.47)	865 (5.95%)	10.78 (10.51, 11.06)
45-64	1,548 (10.65%)	11.92 (11.70, 12.13)	1,711 (11.77%)	11.67 (11.49, 11.85)
>=65	1,023 (7.04%)	13.49 (13.29, 13.69)	1,109 (7.63%)	12.87 (12.61, 13.14)

Table 4

Weighted mean sedentary time (hours/day) by race/ethnicity and BMI category, NHANES 2011-2014.

Table 8: Weighted mean sedentary time (hours/day) by race/ethnicity and BMI category, NHANES 2011-2014.

Race	Underweight/normal	Overweight	Obesity
All	10.77 (10.66, 10.89)	11.29 (11.14, 11.43)	12.06 (11.88, 12.24)
Mexican American	10.45 (10.20, 10.69)	10.35 (10.06, 10.63)	10.90 (10.66, 11.14)
NH Asian	10.98 (10.73, 11.23)	11.52 (11.18, 11.86)	11.83 (11.22, 12.45)
NH Black	11.07 (10.82, 11.31)	11.65 (11.28, 12.03)	12.42 (12.05, 12.79)
NH White	10.76 (10.59, 10.93)	11.41 (11.21, 11.61)	12.23 (11.98, 12.47)
Other Hispanic	10.47 (10.15, 10.80)	10.59 (10.22, 10.97)	11.10 (10.72, 11.48)
Other/Multi	11.01 (10.59, 11.43)	11.37 (10.30, 12.45)	13.01 (12.43, 13.59)

Figure 4

Sleep-cutoff calibration and sedentary-time sensitivity. Survey-weighted mean accelerometer-derived sleep duration and waking sedentary time are shown across candidate sleep-percentage cutoffs. Shaded bands indicate 95% confidence intervals. The vertical dashed line marks the selected 90.9% cutoff, calibrated to the self-reported sleep-duration benchmark.

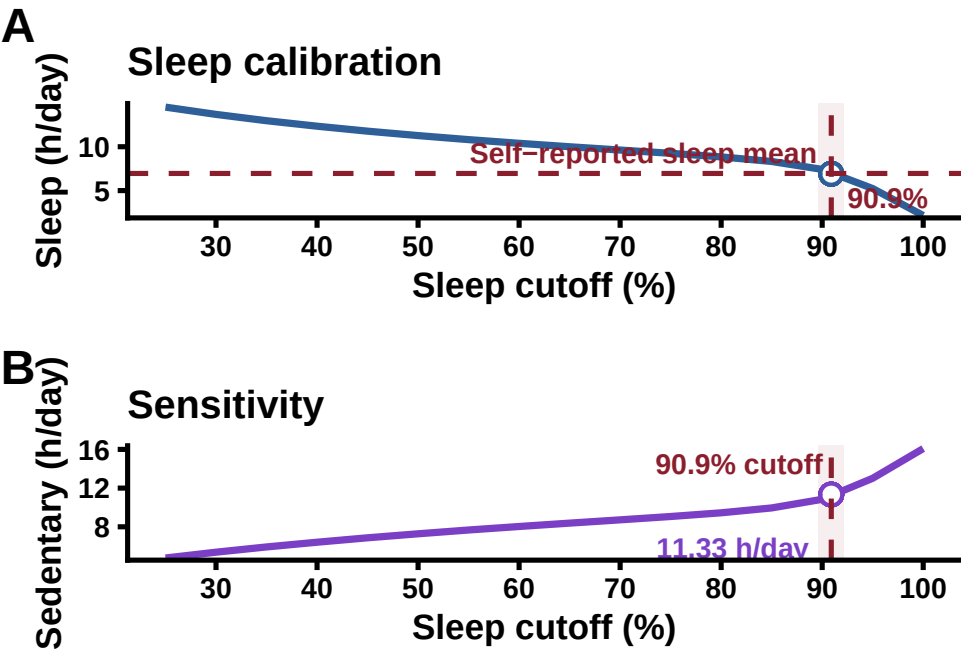


Figure 5: Sleep-cutoff calibration and sedentary-time sensitivity. Survey-weighted mean accelerometer-derived sleep duration and waking sedentary time are shown across candidate sleep-percentage cutoffs. Shaded bands indicate 95% confidence intervals. The vertical dashed line marks the selected 90.9% cutoff, calibrated to the self-reported sleep-duration benchmark.

Output Index

Table 9: Manuscript output index.

Output	Path
Table 1	outputs/tables/table1_wear_day_distribution.csv
Table 2	outputs/tables/table2_study_comparison.csv
Table 3	outputs/tables/table3_age_gender_sedentary.csv
Table 4	outputs/tables/table4_race_bmi_sedentary.csv
Main results	outputs/results/main_results.csv
Main results by cycle	outputs/results/main_results_by_cycle.csv
Demographic missingness	outputs/qc/demographic_missingness.csv
Cutoff sensitivity grid	outputs/cutoff/cutoff_sensitivity_grid.csv
Objective curve	outputs/cutoff/objective_curve.csv
Selected cutoff	outputs/cutoff/selected_cutoff.csv
Analytic participant sedentary data	outputs/datasets/analytic_participant_sedentary.csv
Analytic participant sedentary documentation	outputs/documentation/analytic_participant_sedentary_documentation.txt
Analytic epoch sedentary data	outputs/datasets/analytic_epoch_sedentary.csv
Analytic epoch sedentary documentation	outputs/documentation/analytic_epoch_sedentary_documentation.txt
Figure 1	figures/manuscript/fig1.png
Figure 2	figures/manuscript/figure2_hourly_daytype.png
Figure 3	figures/manuscript/sedentary_distribution_with_24h_composition.png
Figure 4	figures/manuscript/cutoff_sensi.png